

SOCCER INFRARED

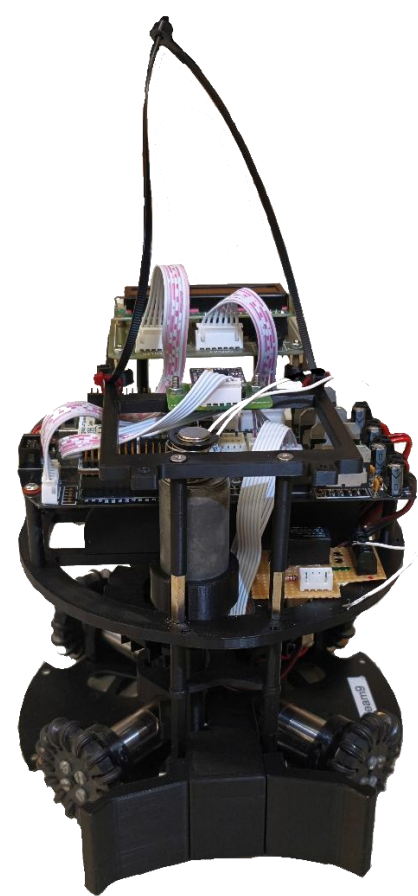
TEAM ArchLinux | BRG Kepler | RoboCup Junior

WHAT IS IT?

- **Autonomous Soccer Robot:** A high-performance robotic system based on the KeplerBRAIN V4 platform, engineered for fully automatic detection, tracking, and precise interception of an infrared-emitting soccer ball.
- **The Mission:** Lightning-fast ball acquisition and dynamic field navigation to strategically outmanoeuvre opposing robotic systems.

HOW IT WORKS

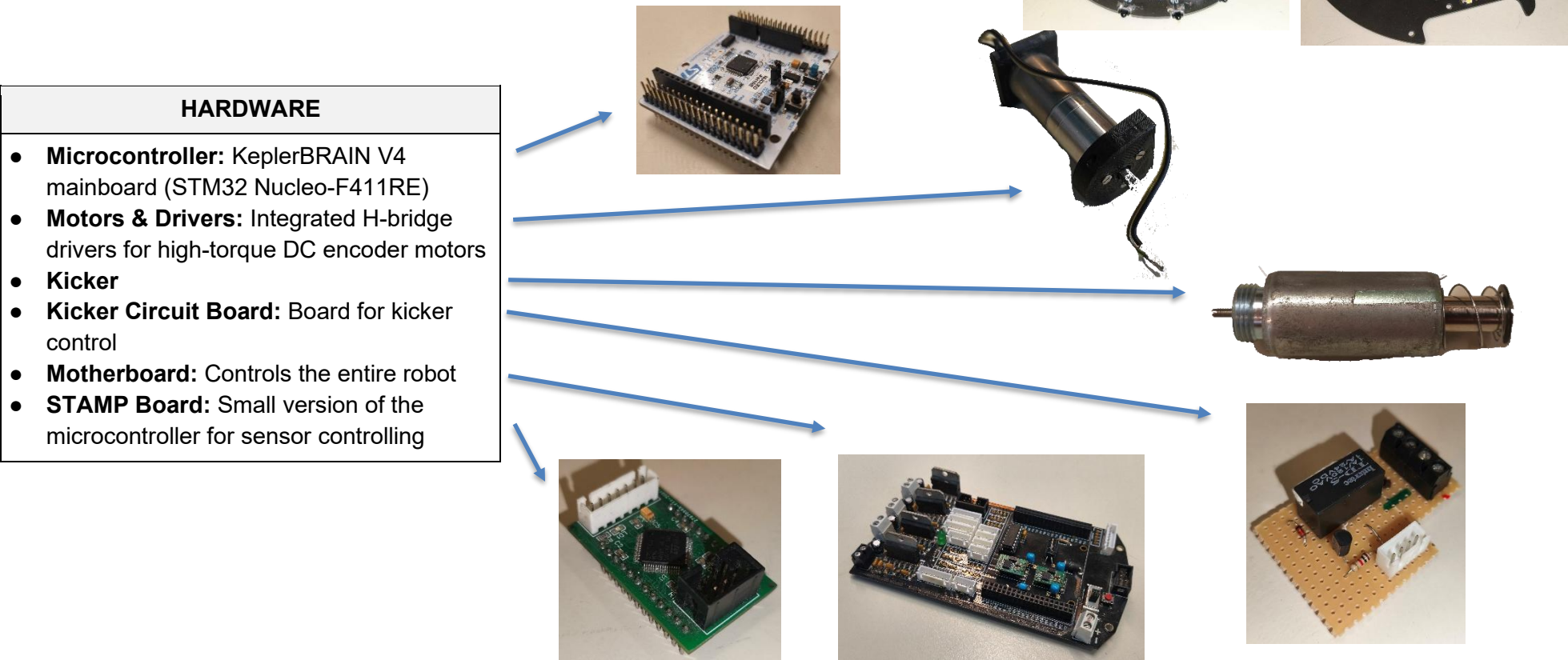
- **IR Ball Detection:** An advanced, multi-channel photodiode array continuously scans the environment and filters out ambient light noise to instantly isolate the pulsed IR signal of the ball.
- **Movement Logic:** High-speed vector calculations translate the relative ball angle directly into active motor commands, allowing the robot to intercept the ball dynamically instead of just chasing it.
- **Key Steps:**
 1. 16 sensors for exact soccer ball tracking.
 2. Exact ball position calculation using vector sums.
 3. Precise PID control loop execution for heading and drift correction.
 4. Transition to attack or dribble mode upon close-range ball acquisition.



TECHNICAL SPECIFICATIONS

SOFTWARE	TEAM	Sensors
• Programming Language: C++ via the Arduino IDE framework (object-oriented). • Control Theory: Closed-loop PID controller utilizing real-time gyro telemetry for heading correction. • Decision Making: Finite State Machine (FSM) for seamless transitions between searching, chasing, and attacking.	• Team Name: ArchLinux • Team members: Sophia Juffinger, Felix Pfragner, Maximilian Scholz, Julian Wendler • School: BRG Kepler, Graz • Platform: KeplerBRAIN V4 Ecosystem	• IMU-Sensor • Ground / Line Sensor • IR-Sensor array

HARDWARE
• Microcontroller: KeplerBRAIN V4 mainboard (STM32 Nucleo-F411RE) • Motors & Drivers: Integrated H-bridge drivers for high-torque DC encoder motors • Kicker • Kicker Circuit Board: Board for kicker control • Motherboard: Controls the entire robot • STAMP Board: Small version of the microcontroller for sensor controlling



TESTS / RESULTS / IMPROVEMENTS

- **Performance:** Successfully achieved reliable ball tracking and stable, straight-line acceleration across the entire field.
- **Challenges:** Unfiltered raw IR values initially caused jittery movements and sensor blind spots near the chassis edges during rapid turns.
- **Optimizations:** Implemented a rolling average software filter for signal smoothing, integrated IMU telemetry for orientation, and optimized turning algorithms to maintain a constant ball lock.